

PG_DB

postgres

InfluxDB

INFLUX_URL

http://localhost:8086

INFLUX_TOKEN

lUDz6ChzF0UdOi-11lc6w_rQs2Z29WT-isCDn4/Dh7Ced7aV1

INFLUX_ORG

UIT

INFLUX_BUCKET

DistributedLogging

Influx Docker container

csdInc-influxdb-1

Run config

TOTAL_ROWS

1000000

BATCH_SIZE

1000

INFLUX_CHUNK

100

Verify every N batches

20

☒ Xóa data cũ trước khi insert

☒ Ghi log ra file

OUTPUT_FILE

/Users/henry/Documents

Run benchmark

3DB Distributed Logging Benchmark ↔

Click Run benchmark để generate dữ liệu, insert vào ClickHouse, TimescaleDB, InfluxDB, rồi hiển thị log, bảng thống kê và biểu đồ.

960,000 / 1,000,000 rows / 01:15:52 (hh:mm:ss)

Batch 945 | 945,000 / 1,000,000 | 94.5% | 306 rows/s

Batch 946 | 946,000 / 1,000,000 | 94.6% | 306 rows/s

Batch 947 | 947,000 / 1,000,000 | 94.7% | 306 rows/s

Batch 948 | 948,000 / 1,000,000 | 94.8% | 306 rows/s

Batch 949 | 949,000 / 1,000,000 | 94.9% | 306 rows/s

Batch 950 | 950,000 / 1,000,000 | 95.0% | 305 rows/s

Batch 951 | 951,000 / 1,000,000 | 95.1% | 305 rows/s

Batch 952 | 952,000 / 1,000,000 | 95.2% | 305 rows/s

Batch 953 | 953,000 / 1,000,000 | 95.3% | 305 rows/s

Batch 954 | 954,000 / 1,000,000 | 95.4% | 305 rows/s

Batch 955 | 955,000 / 1,000,000 | 95.5% | 305 rows/s

Batch 956 | 956,000 / 1,000,000 | 95.6% | 305 rows/s

Batch 957 | 957,000 / 1,000,000 | 95.7% | 304 rows/s

Batch 958 | 958,000 / 1,000,000 | 95.8% | 304 rows/s

Batch 959 | 959,000 / 1,000,000 | 95.9% | 304 rows/s

Batch 960 | 960,000 / 1,000,000 | 96.0% | 304 rows/s

Verify batch 960

Point lookup id: 4425a887-102a-41d9-9330-f6cf5aebc898

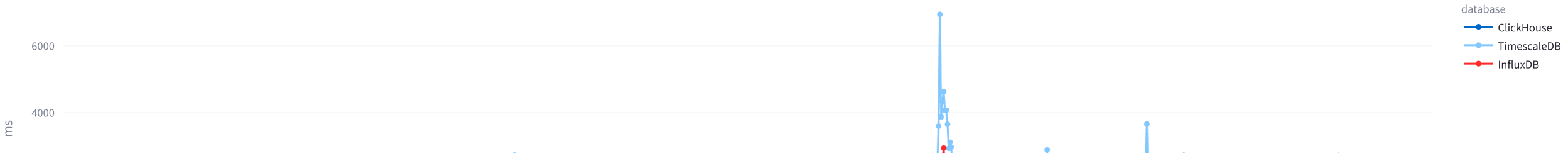
Trace lookup trace_id: tr-c645016dcd35cd8ee0eac0ae9a990218

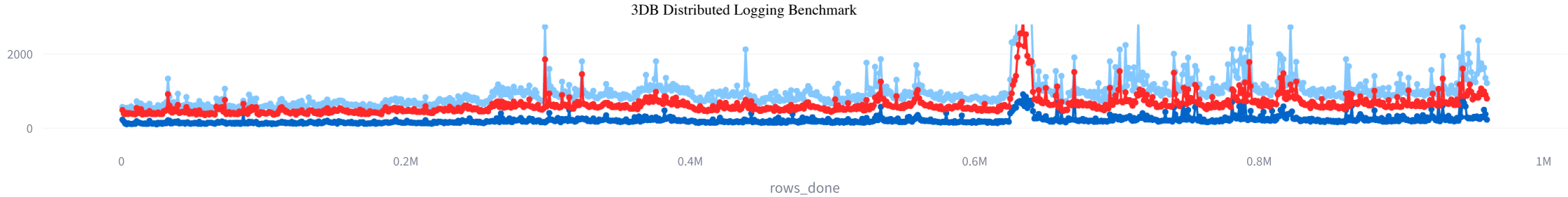
Free-text sample: services from unexpected traffic spikes.

Insert 1,000 rows time

batch	batch_rows	rows_done	ClickHouse_ms	TimescaleDB_ms	InfluxDB_ms
86	1000	86000	404.84	738.97	657.51
87	1000	87000	128.59	638.6	482.71
88	1000	88000	144.59	544.12	402.95
89	1000	89000	126.34	529.21	384.96
90	1000	90000	178.11	905.39	528.09
91	1000	91000	142.23	661.88	467.91
92	1000	92000	168.55	790.19	568.74
93	1000	93000	136.53	675.84	500.53
94	1000	94000	149.44	776.78	526.57
95	1000	95000	170.51	801.63	563.29
96	1000	96000	121.04	533.45	388.05

Insert 1,000 rows time





Storage size

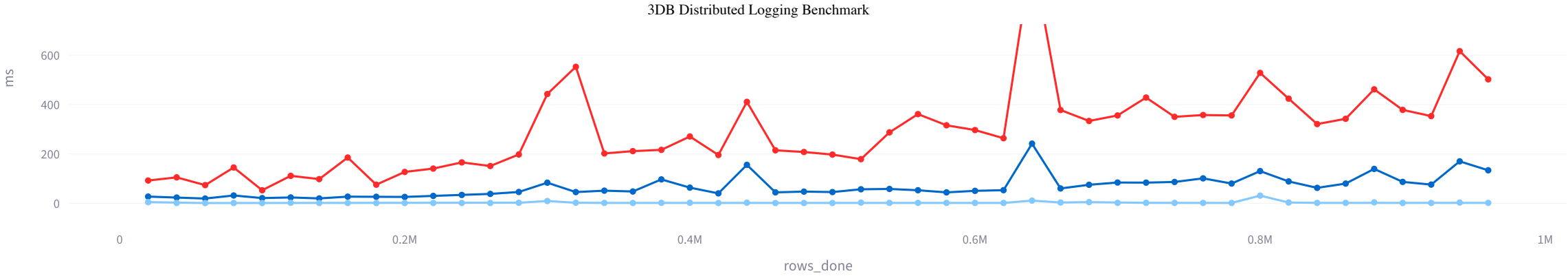
batch	rows_done	database	size
220	220000	TimescaleDB	938 MB
220	220000	InfluxDB	434M
240	240000	ClickHouse	136.34 MiB
240	240000	TimescaleDB	1024 MB
240	240000	InfluxDB	472M
260	260000	ClickHouse	147.71 MiB
260	260000	TimescaleDB	1110 MB
260	260000	InfluxDB	510M
280	280000	ClickHouse	159.03 MiB
280	280000	TimescaleDB	1196 MB
280	280000	InfluxDB	546M

Point lookup

batch	rows_done	database	ms	result
20	20000	ClickHouse	27.2951	1
20	20000	TimescaleDB	4.9622	1
20	20000	InfluxDB	92.411	1
40	40000	ClickHouse	23.4511	1
40	40000	TimescaleDB	1.5149	1
40	40000	InfluxDB	105.1879	1
60	60000	ClickHouse	19.4278	1
60	60000	TimescaleDB	1.1091	1
60	60000	InfluxDB	73.7591	1
80	80000	ClickHouse	31.8773	1

Point lookup time

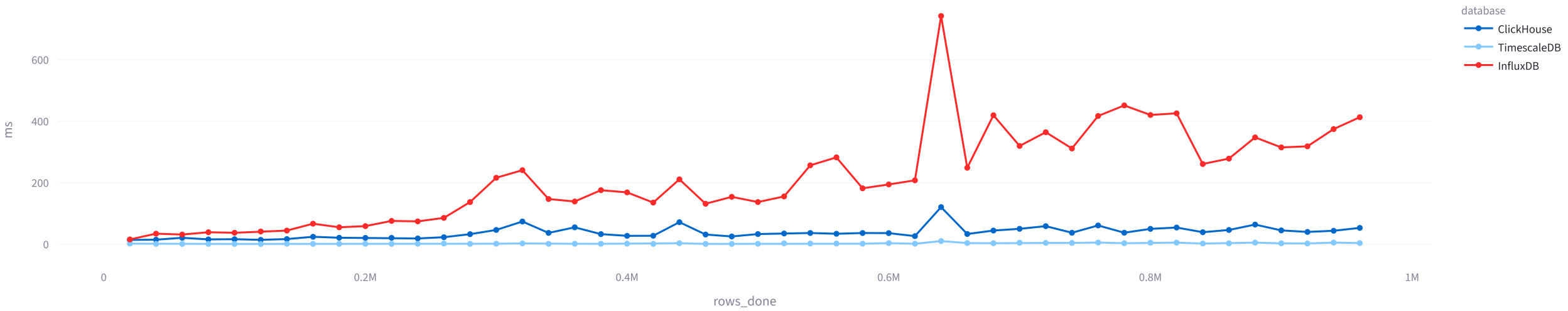




Trace lookup

batch	rows_done	database	ms	result
20	20000	ClickHouse	14.0781	11
20	20000	TimescaleDB	2.044	11
20	20000	InfluxDB	15.6732	11
40	40000	ClickHouse	14.7982	20
40	40000	TimescaleDB	1.152	20
40	40000	InfluxDB	34.3421	20
60	60000	ClickHouse	20.9241	38
60	60000	TimescaleDB	0.963	38
60	60000	InfluxDB	31.5228	38
80	80000	ClickHouse	15.7237	27

Trace lookup time

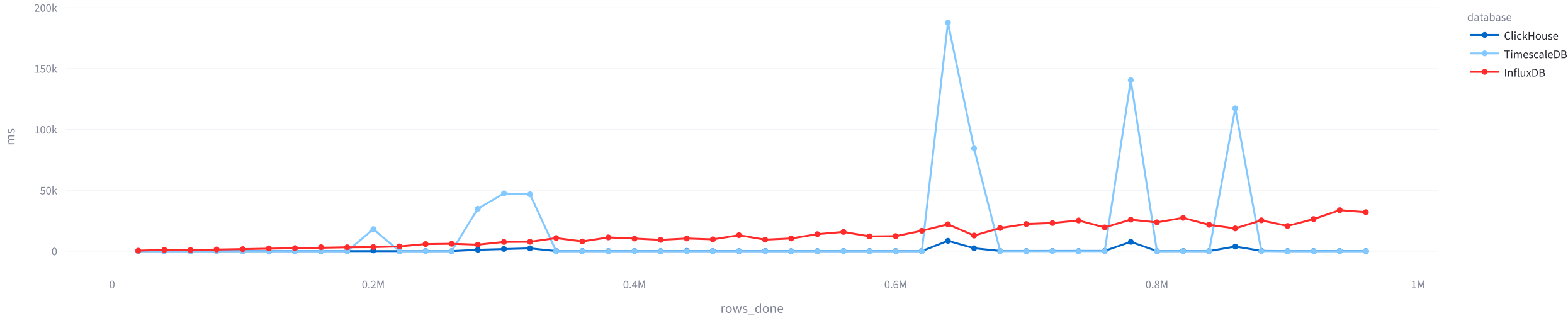


Free-text search

3DB Distributed Logging Benchmark

batch	rows_done	database	ms	result
	700	InfluxDB	22307.95	1767
	720	ClickHouse	76.3929	10
	720	TimescaleDB	22.3451	10
	720	InfluxDB	23129.878	1745
	740	ClickHouse	62.701	10
	740	TimescaleDB	10.8459	10
	740	InfluxDB	25223.4402	1744
	760	ClickHouse	216.8629	10
	760	TimescaleDB	21.6591	10
	760	InfluxDB	19518.4879	1749

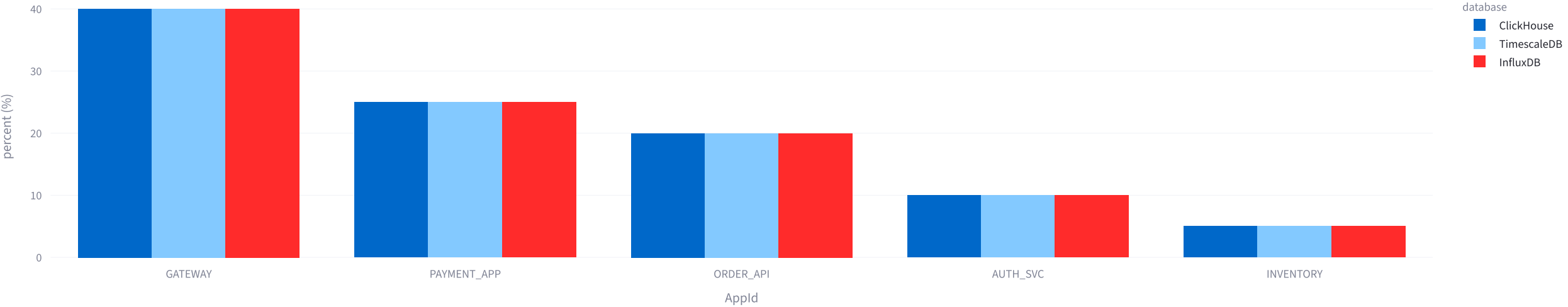
Free-text search time



AppId distribution

database	label	count	percent
ClickHouse	GATEWAY	384274	40.03
ClickHouse	PAYMENT_APP	239911	24.99
ClickHouse	ORDER_API	191697	19.97
ClickHouse	AUTH_SVC	96163	10.02
ClickHouse	INVENTORY	47955	5
TimescaleDB	GATEWAY	384274	40.03
TimescaleDB	PAYMENT_APP	239911	24.99
TimescaleDB	ORDER_API	191697	19.97
TimescaleDB	AUTH_SVC	96163	10.02
TimescaleDB	INVENTORY	47955	5

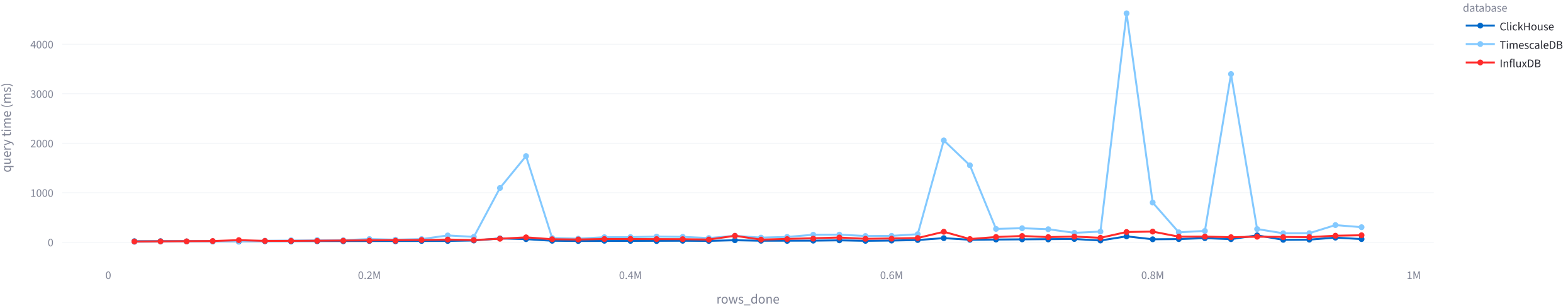
AppId distribution



AppId distribution query time

batch	rows_done	database	ms	status
20	20000	ClickHouse	19.39	OK
20	20000	TimescaleDB	6.49	OK
20	20000	InfluxDB	13.39	OK
40	40000	ClickHouse	19.82	OK
40	40000	TimescaleDB	11.74	OK
40	40000	InfluxDB	16.28	OK
60	60000	ClickHouse	15.33	OK
60	60000	TimescaleDB	14.66	OK
60	60000	InfluxDB	17.08	OK
80	80000	ClickHouse	14.83	OK

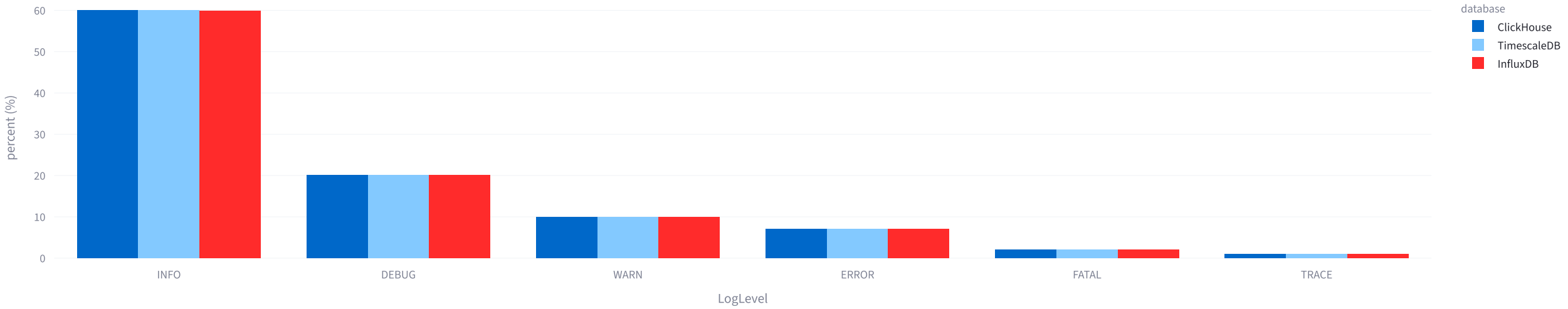
AppId distribution query time



LogLevel distribution

database	label	count	percent
TimescaleDB	WARN	95728	9.97
TimescaleDB	ERROR	67255	7.01
TimescaleDB	FATAL	19164	2
TimescaleDB	TRACE	9602	1
InfluxDB	INFO	574733	59.95
InfluxDB	DEBUG	192267	20.06
InfluxDB	WARN	95688	9.98
InfluxDB	ERROR	67235	7.01
InfluxDB	FATAL	19163	2
InfluxDB	TRACE	9601	1

LogLevel distribution

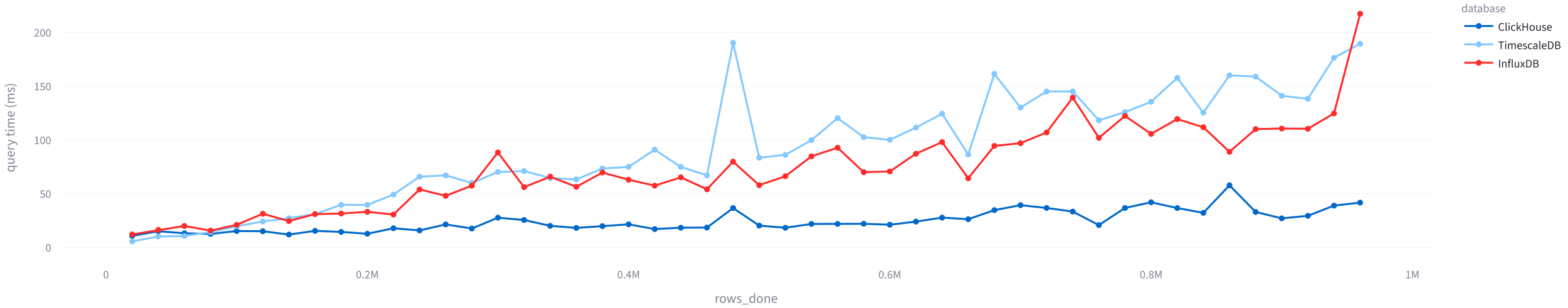


LogLevel distribution query time

3DB Distributed Logging Benchmark

batch	rows_done	database	ms	status
900	900000	InfluxDB	110.64	OK
920	920000	ClickHouse	29.43	OK
920	920000	TimescaleDB	138.31	OK
920	920000	InfluxDB	110.43	OK
940	940000	ClickHouse	38.95	OK
940	940000	TimescaleDB	176.51	OK
940	940000	InfluxDB	124.73	OK
960	960000	ClickHouse	41.72	OK
960	960000	TimescaleDB	189.51	OK
960	960000	InfluxDB	217.41	OK

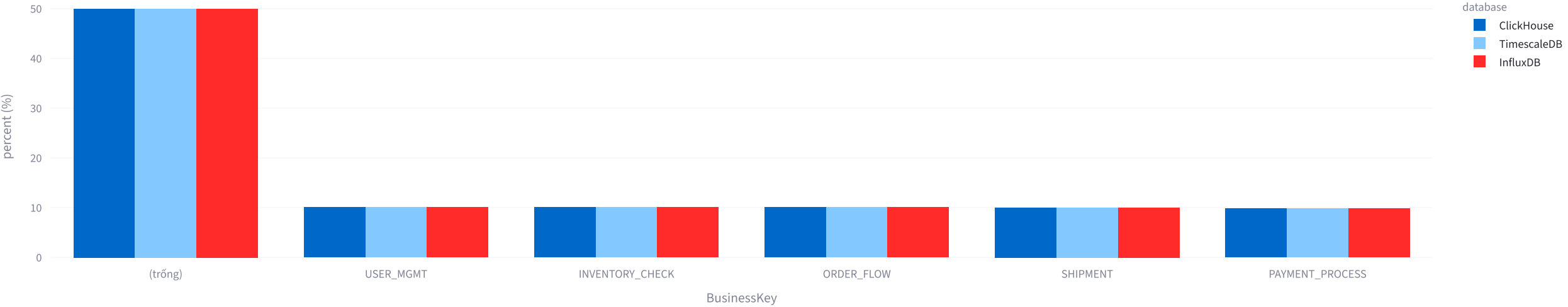
LogLevel distribution query time



BusinessKey distribution

database	label	count	percent
TimescaleDB	INVENTORY_CHECK	96701	10.07
TimescaleDB	ORDER_FLOW	96270	10.03
TimescaleDB	SHIPMENT	96112	10.01
TimescaleDB	PAYMENT_PROCESS	94469	9.84
InfluxDB	(trống)	478993	49.96
InfluxDB	USER_MGMT	96603	10.08
InfluxDB	INVENTORY_CHECK	96578	10.07
InfluxDB	ORDER_FLOW	96145	10.03
InfluxDB	SHIPMENT	95994	10.01
InfluxDB	PAYMENT_PROCESS	94374	9.84

BusinessKey distribution



BusinessKey distribution query time

batch	rows_done	database	ms	status
20	20000	ClickHouse	11.89	OK
20	20000	TimescaleDB	6.41	OK
20	20000	InfluxDB	11.27	OK
40	40000	ClickHouse	40.08	OK
40	40000	TimescaleDB	13.56	OK
40	40000	InfluxDB	21.75	OK
60	60000	ClickHouse	16.28	OK
60	60000	TimescaleDB	16.47	OK
60	60000	InfluxDB	16.45	OK
80	80000	ClickHouse	13.22	OK

BusinessKey distribution query time

